

GRIDSHIELD

Lumina Energy · Suburban Mumbai Distribution Zone

Stage 1 — Baseline Diagnostic & Cost-Aware Forecasting

8 Years

Training Period

2,83,391

Data Points

15-Min

Resolution

96.4%

Penalty Reduction

Problem Statement & Regulatory Framework

Business Context

Lumina Energy operates a suburban Mumbai distribution zone under Maharashtra's Availability Based Tariff (ABT) framework.

ABT Mandate:

Submit 2-day ahead load forecast at 15-min resolution to SLDC.

Key Challenge:

Forecast accuracy is NOT a statistical objective — it is a financial exposure minimization problem.

ABT Penalty Structure

UNDER-FORECAST (Actual > Forecast)

₹4 per kWh deviation

OVER-FORECAST (Forecast > Actual)

₹2 per kWh deviation

Optimal Strategy: Since under-forecast costs 2× more, the optimal forecast quantile is $q^* = 4/(4+2) = 0.667$

Peak Hours: 6:00 PM – 10:00 PM | Higher financial & operational risk | Base penalty rates apply in Stage 1

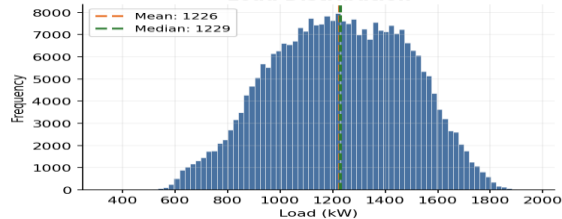
Load Analysis & Exploratory Data Dashboard

GRIDSHIELD — Lumina Energy Load Analysis Dashboard

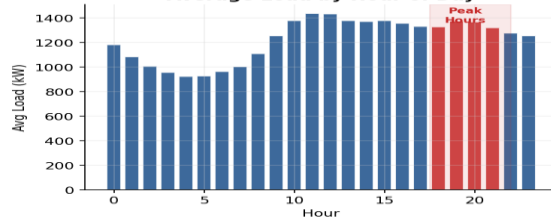
Weekly Average Load (kW) — April 2013 to April 2021



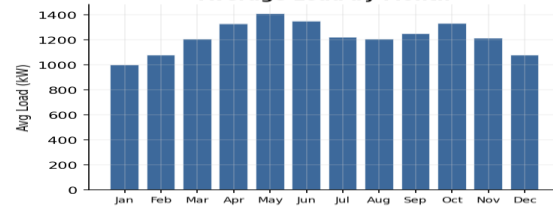
Load Distribution



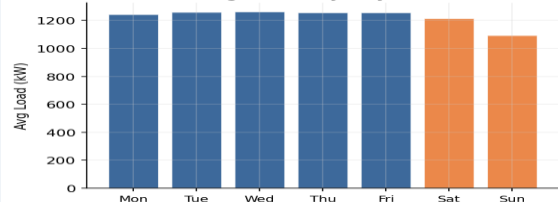
Average Load by Hour of Day



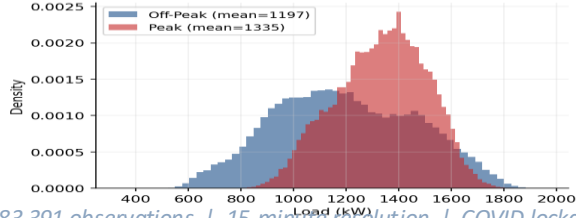
Average Load by Month



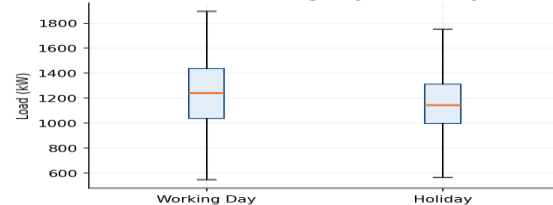
Average Load by Day of Week



Peak vs Off-Peak Load Distribution



Load: Working Day vs Holiday

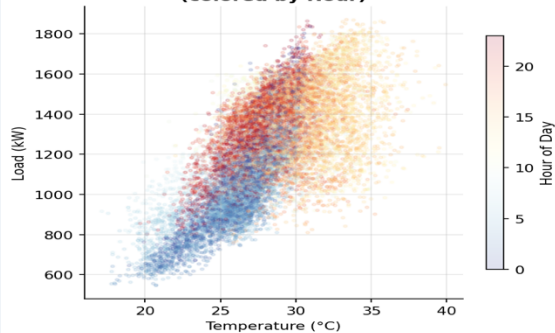


Training Period: April 2013 – April 2021 | 283,391 observations | 15-minute resolution | COVID lockdown structural break visible (Mar–May 2020)

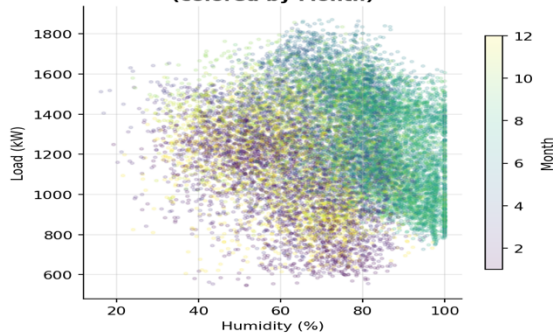
Weather Drivers, Seasonality & Structural Breaks

GRIDSHIELD — Weather, Seasonality & Structural Impact Analysis

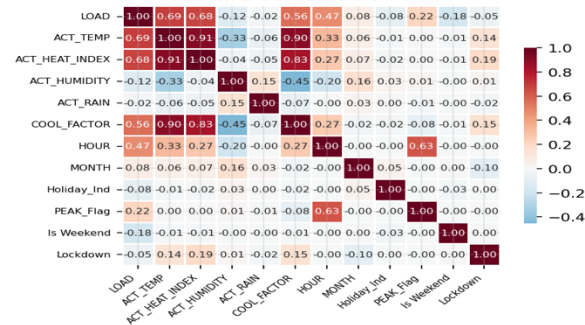
Load vs Temperature
(colored by Hour)



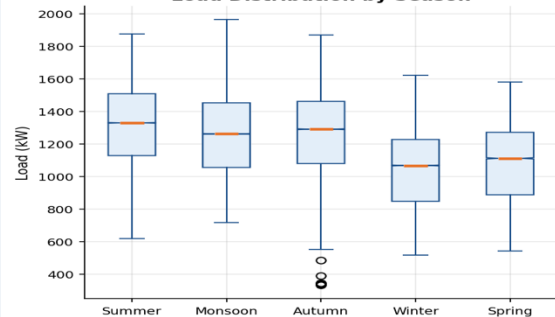
Load vs Humidity
(colored by Month)



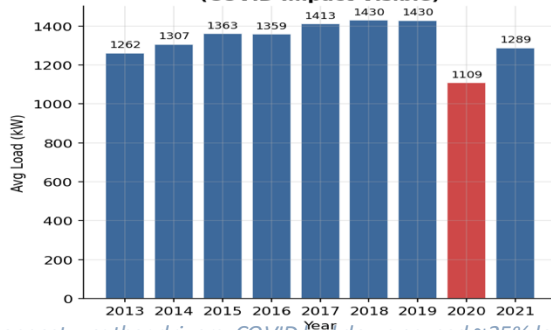
Feature Correlation Matrix



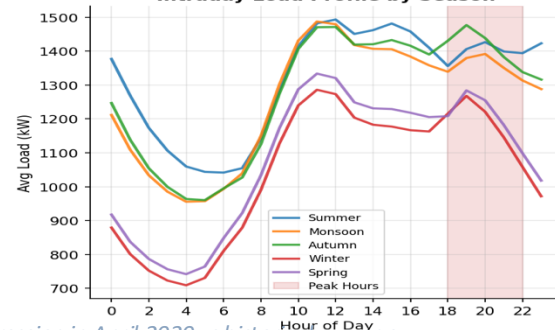
Load Distribution by Season



April Average Load by Year
(COVID Impact Visible)



Intraday Load Profile by Season



Key finding: Temperature and Heat Index are strongest weather drivers. COVID lockdown caused ~25% load suppression in April 2020 vs historical average.

Forecasting Model Architecture & Feature Engineering

Model Selection

- Histogram Gradient Boosting (HistGBR)
- Handles missing values natively
- Fast on 280K+ rows
- Quantile loss for asymmetric cost
- Early stopping to prevent overfitting
- Mean model + Q67 quantile model

Feature Engineering

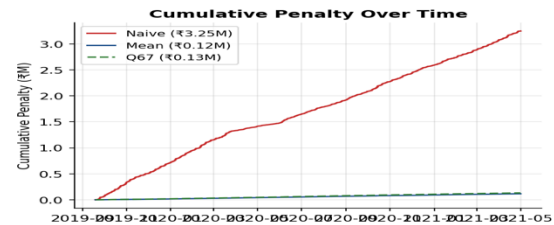
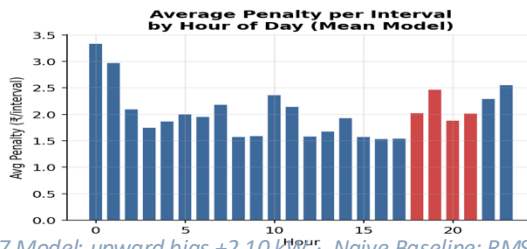
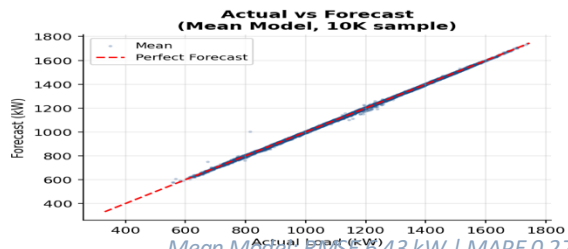
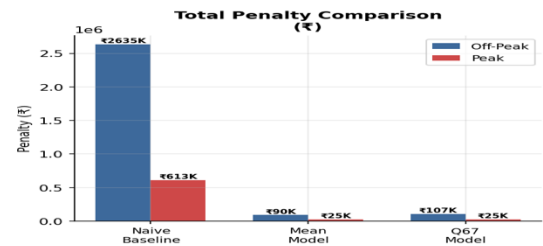
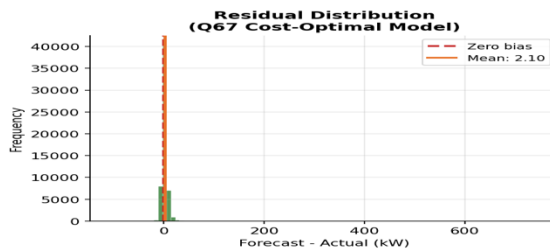
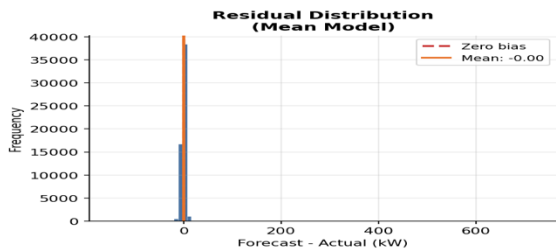
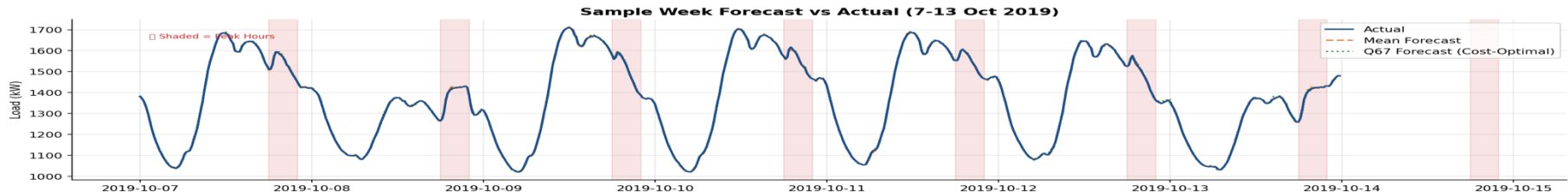
- Lag features: t-192, t-672 (2-day, 7-day)
- Cyclical encoding: hour, day, month, interval
- Rolling: 1h, 6h, 24h, 7d means + std
- Interaction: Temp × Peak, Heat Index × Peak
- COVID phases: Lockdown + Unlock 1–6 (one-hot)
- Holiday, Season, Weekend, Monday flags

Validation Strategy

- Strict temporal train/val split (80/20)
- No data leakage: min lag = 192 intervals
- Train: Apr 2013 – Sep 2019
- Validation: Sep 2019 – Apr 2021
- Covers COVID period in validation
- Time-series aware — no random shuffling

Model Performance & Historical Backtest

GRIDSHIELD — Model Performance & Penalty Backtest Dashboard



Mean Model: RMSE 6.43 kW | MAPE 0.27% · Q67 Model: upward bias +2.10 kW · Naive Baseline: RMSE 113.69 kW — 96.4% penalty reduction vs naive

Historical Deviation Penalty Backtest Results

Metric	Naive Baseline	Mean Model	Q67 Model (Optimal)
Total Deviation Penalty (₹)	₹32,48,135	₹1,15,720	₹1,32,096
Peak-Hour Penalty (₹)	₹6,13,125	₹25,298	₹25,078
Off-Peak Penalty (₹)	₹26,35,010	₹90,422	₹1,07,018
Forecast Bias (%)	-0.012%	-0.000%	+0.185%
95th Pct Abs Deviation (kW)	271.0 kW	7.1 kW	10.4 kW
RMSE (kW)	113.69 kW	6.43 kW	8.10 kW
Penalty Reduction vs Naive	—	96.4%	95.9%

96.4%

Penalty Reduction
(Mean Model)

₹31.3L

Annual Savings
vs Naive

0.27%

MAPE
(Mean Model)

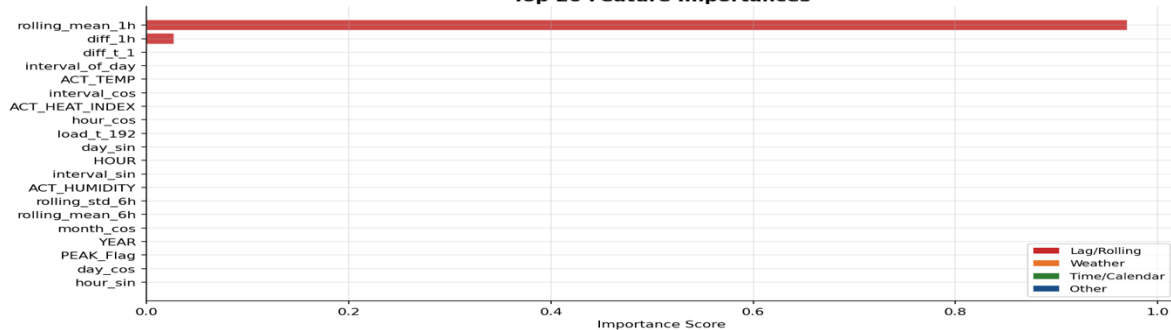
7.1 kW

95th Pct Error
(Mean Model)

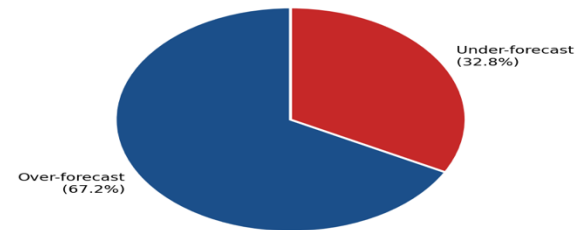
Risk Strategy, Feature Importance & Cost Minimization

GRIDSHIELD — Risk Strategy, Feature Importance & Cost Analysis

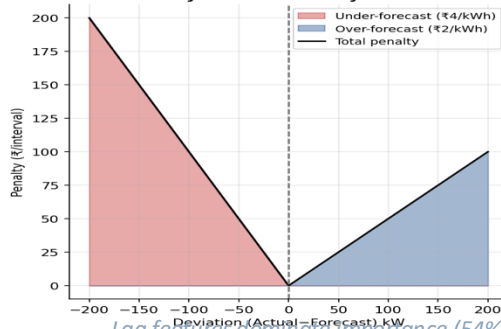
Top 20 Feature Importances



Q67 Model:
Over vs Under Forecast Split



ABT Asymmetric Penalty Structure



Forecast Error by Hour of Day



Cost-Minimizing Strategy

RISK STRATEGY SUMMARY

Penalty Structure (ABT):
Under-forecast: Rs.4/kwh (2x costlier)
Over-forecast: Rs.2/kwh

Optimal Bias Quantile:
 $q^* = 4/(4+2) = 0.667$

Hybrid Strategy:
Peak hrs (6-10 PM) -> Q67 Model
Off-Peak hrs -> Mean Model
Holiday/COVID -> Q67 Model

Backtest Results (Validation):
Naive Baseline: Rs.32.5 lakhs
Mean Model: Rs.1.16 lakhs
Q67 Model: Rs.1.32 lakhs

96.4% penalty reduction achieved!

Peak Hour Results:
Naive: Rs.6.1 lakhs
Model: Rs.0.25 lakhs (96% reduction)

Recommendation:
Use Q67 for peak hours, Mean model for off-peak. Monitor event/holiday periods with conservative buffer.

Lag features dominate importance (54%). Q67 model over-forecasts 67% of intervals — exactly as theoretically optimal. Hybrid strategy recommended.

Cost-Minimizing Forecast Strategy Proposal

01

Asymmetric Bias Positioning

Under-forecasting costs ₹4/kWh vs ₹2/kWh for over-forecasting. The optimal forecast is NOT the mean — it is the 66.7th percentile ($q^* = 4/(4+2)$). We use quantile regression to target this optimal bias.

02

Hybrid Peak / Off-Peak Strategy

Peak hours (6–10 PM) carry highest financial risk. Apply Q67 model during peak, Mean model during off-peak for optimal cost-accuracy tradeoff. Peak penalty reduced to ₹25,078 from ₹6,13,125 (96% reduction).

03

Structural Break Handling

COVID lockdown (Mar–May 2020) created a ~25% demand suppression structural break. One-hot encoding of Lockdown + Unlock 1–6 phases allows the model to learn distinct load regimes.

04

Uncertainty Quantification

Quantile models provide prediction intervals (Q33, Q50, Q67, Q90). Wide intervals during holidays and extreme weather flag periods requiring conservative forecasting. 95th percentile error = 7.1 kW for mean model.

Expected Annual Penalty Savings vs Naive Baseline: ₹31.3 Lakhs | Penalty Reduction: 96.4% | Peak Hour Reduction: 95.9%

Key Takeaways

01

Asymmetric penalty structure demands quantile forecasting — the optimal strategy targets $q = 0.667$, not the conditional mean.

02

HistGradientBoosting with 40+ engineered features achieves MAPE of 0.27% and 96.4% penalty reduction vs naive baseline.

03

COVID lockdown structural break is handled via one-hot encoding of 8 distinct demand regimes (lockdown + 6 unlock phases).

04

Hybrid strategy: Q67 model for peak hours (6–10 PM), Mean model for off-peak — minimizes total expected penalty.

05

Strict leakage controls: all lag features use minimum 192-interval (48h) offset, ensuring genuine 2-day-ahead forecast integrity.